

MLFF Architecture Revision 47

PERF-P2 lazy TARGET-DATA2C authority v2

mdstats project

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Revision 47 closes PERF-P2 in mdstats 0.20.181a0 and makes TargetDataLadderPlan.v2 the current TARGET-DATA2C authority.

The ladder now materializes configured rungs in ascending order and stops only after the active TARGET-DATA2D shortlist width has been satisfied by the same global coverage-plus-mandatory predicate used by Stage A. The canonical width is four, but noncanonical configured widths are preserved exactly.

v2 records the configured sequence, exact materialized prefix, global rung qualification records, early-stop qualifiers, intentionally skipped sizes, pool-unavailable sizes, monotonicity contract, last materialized rung, and stop reason. Legacy v1 remains readable as a qualification oracle but is stale for current campaign authority.

Qualification against exhaustive v1 preserves Stage-A survivor sizes, survivor membership, coverage-report digests, mandatory status, and the TARGET-DATA2D shortlist. A 37,633-frame supplied-data-derived forced early-stop fixture reduces median wall time from 7.867 s to 1.556 s and serialized ladder authority from 4,729,481 B to 591,058 B. The exhaustive fallback path remains exact; its noisy timing is not promoted as a speed claim.

PERF-P3 - CPU structural and reduction hardening - is next.